

Quiz #2 (10/21/2025)

Name: Diana RODRIGUEZ

We are separating a mixture of ethane (C₂), propane (C₃), n-butane (C₄), and n-pentane (C₅) in a distillation column operating at 5.0 atm. The column has a total condenser and a partial reboiler. The feed flow rate is 1000 kmol/h. The feed is a saturated liquid. Feed is 10 mol% ethane, 30 mol% propane, 50 mol% n-butane, and 10 mol% n-pentane. K values are K_{C2}=4.6, K_{C3}=1.37, K_{C4}=0.35, and K_{C5}=0.10. A 99.5% recovery of propane is desired in the distillate. A 99.0% recovery of n-butane is desired in the bottoms.

$$LK = \int_{C_3} = 99.5 \quad \int_{C_4} = 99.0 = HK$$

- 1) (2pts) Identify each component such as LNK, LK, NK, HK, or HNK

$$C_3 = LK \quad C_4 = HK \quad C_2 = LNK \quad C_5 = HNK$$

- 2) (2pts) Write down fractional recovery (e.g., $FR_{i,D} = x_{i,D} \cdot D / F \cdot z_i$) of ethane, propane, n-butane, and n-pentane both in distillate and bottom

	Distillate	Bottom
Ethane C ₂	0	1
Propane C ₃	0.995	0.005
Butane C ₄	0.01	0.99
Pentane C ₅	0	1

- 3) (3pts) Calculate the distillate flow rate (D).

$$\int_{x_i} = \frac{D x_{i,D}}{F \cdot z_i}$$

$$D x_{C_4} = 500 \quad D x_{C_3} = 1000 \quad D x_{C_2} = 298.5$$

$$D = 403.5 \frac{\text{kmol}}{\text{hr}}$$

$$C_2 \Rightarrow D x_{C_2} = 1000 \frac{\text{kmol}}{\text{hr}} (0.10)$$

$$C_3 \Rightarrow D x_{C_3} = 1000 \frac{\text{kmol}}{\text{hr}} (0.30) (0.995)$$

$$C_4 \Rightarrow D x_{C_4} = 1000 (0.50) (0.01) (0.99 - 1)$$

$$C_5 \Rightarrow D x_{C_5} = 1000 (0.10) (0)$$

- 4) (3pts) Find $N_{\min}$ from the Fenske equation (hint: use the relationship between α and K value)

$$D x_{C_5} = 0 \frac{\text{kmol}}{\text{hr}}$$

$$N_{\min} = \frac{\ln \left[\frac{(FR_A)_{\text{dist}} (FR_B)_{\text{bot}}}{[1 - (FR_A)_{\text{dist}}][1 - (FR_B)_{\text{bot}}]} \right]}{\ln \alpha_{AB}}$$

where, $(FR_A)_{\text{dist}}$ = fractional recovery of LK in distillate (i.e., A=LK)
 $(FR_B)_{\text{bot}}$ = fractional recovery of HK in distillate or bottom (i.e., B=HK)
 α_{AB} = the relative volatility

$$C = LNK$$

$$N_{\min} = \frac{\ln \left\{ \frac{(0.995)(0.99)}{(1-0.995)(1-0.99)} \right\}}{\ln \left(\frac{1.37}{0.35} \right)}$$

$$K_{C_3} = 1.37 \quad K_{C_4} = 0.35$$

$$\frac{LK}{HK} = \alpha_{AB}$$

$$\ln \left(\frac{0.995}{0.005 \cdot 0.01} \right)$$

$$9850.5$$

$$N_{\min} = 7.2$$